# Real-World Coordinate-Plane Problems; Perimeter

*Teacher's Edition*

## The Period

**MA.8.GR.1.2** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

**Must** Find the perimeter of a figure on the coordinate plane, each slanted side by its right triangle (a triangle with no right angle split by its height); the straight-line and along-the-grid distances on a map, and how much shorter the straight line is; an irrational total kept exact and rounded once.

**Must not** The distance formula (Clarification 2); radicands over 225 or with a square factor; combining radicals (√34 + √34 is two lengths added, not 2√34 — Algebra 1).

**Target** I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

**MTR** **MTR.2.1** — Notes I and III: one perimeter as sides on the grid, as right triangles and as a sum · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.7.1** — Notes II and boards 5, 7 and 8: maps, the streets against the straight line

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. A perimeter adds every side; a slanted side is found by its own right triangle.
2. On a map, the streets add the legs and the straight line is the hypotenuse.
3. No right angle? The height makes two. Round once, at the end.

## Slide by Slide

1. **Real-World Coordinate-Plane Problems; Perimeter.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's idea with only straight sides: a perimeter adds every side — 7, 3, 7 and 3, across zero for both. *1 retrieves 6.05 — a slanted distance by its triangle (yesterday). 2 retrieves 6.03 — a hypotenuse from the legs (√53: a whole radicand up to 225, GR.1.1 Clarification 3) (earlier this week). 3 retrieves 5.04 — a real distance from a map's scale (last unit). 4 retrieves MA.6.GR.1.3 — the perimeter of a rectangle on the coordinate plane (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 10 units 2. √53 ≈ 7.3 m 3. 75 mi 4. 20 units
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Every side counts. A slanted side gets its triangle.
6. **Notes.** The streets add the legs. The straight line is the hypotenuse.
7. **Notes.** No square corner? Make two. Round once, at the end.
8. **Example 1 — A perimeter on the grid.** Ask which sides lie on grid lines before anyone measures: DE and EF.
9. **Example 1 — Worked.** DE: 3 − (−3) = 6; EF: 6 − (−2) = 8; DF: 36 + 64 = 100, 10.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Across 4 − (−5) = 9 and up 5 − (−7) = 12, so the slanted side is √225 = 15 and P = 9 + 12 + 15 = 36 — a board with 21 left the slanted side out.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A map: the streets and the straight line.** Ask what 'along the streets' means on this map: on the grid lines — across, then up.
13. **Example 2 — Worked.** Across 4 − (−3) = 7, up 3 − (−2) = 5.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Across −2 − (−5) = 3 and down 4 − (−4) = 8: by the streets 11 blocks, straight √73 ≈ 8.54, so 8.5 — a board with 11 for the straight line added the legs.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$. Find its perimeter. → ***30 units*** *17 — left the slanted side out [Notes I: every side]; 34 — took AC as 12 + 5 [MA.8.GR.1.2: the theorem, not a sum]*
17. **Board 2.** A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$. Find its perimeter, to the nearest tenth. → ***22.8 units*** *13 — left the slanted side out [Notes I]; 22.9 — rounded 9.849 to 9.85 and then to 9.9 [one rounding, at the end — Notes III]; 26 — the slanted side as 4 + 9 [MA.8.GR.1.2]*
18. **Board 3.** Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle. Find its perimeter. → ***42 units*** *40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]; a perimeter with 12 in it — the height is inside the triangle, not a side [B1G-M, GR.1.2: the theorem on a triangle with no right angle]*

**22–23. Board 4.** What is the perimeter of the triangle with corners (−2, 3), (2, 3) and (2, −4), to the nearest tenth? → ***D — 19.1 units*** *(A) 4 + 7: left the slanted side out [Notes I: every side]; (B) 4 + 7 + 11: the slanted side taken as the sum of the legs [MA.8.GR.1.2: the theorem, not a sum]; (C) 4 + 7 + 65: forgot the square root [B1G-M, GR.1.1: root errors] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** On a map in blocks, the library is at $(-7, -1)$ and the pool at $(5, 4)$. How much shorter is the straight line than the streets? → ***4 blocks shorter*** *13 or 17 — one path, not how much shorter [the question asks a difference]; 30 — added the two paths [the question asks a difference]; 2 across, from 7 − 5 [Florida B1G-M, GR.1.2: integer errors across an axis]*

**26–27. Board 6.** How long is a diagonal of the rectangle with corners (−3, −4), (6, −4), (6, 8) and (−3, 8)? → ***A — 15 units*** *(B) 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]; (C) 9 + 12 + 9 + 12: the perimeter, not one diagonal [Notes II: the straight line is one segment]; (D) 6 and 8, one corner's coordinates, taken as the legs [Math Nation names wrong horizontal and vertical lengths] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$. How far is it straight across, to the nearest tenth? → ***√106 ≈ 10.3 m*** *14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]; √26 — 4 − 5 = 1 for the across leg [Florida B1G-M, GR.1.2: integer errors across an axis]*

**30–31. Board 8.** On a trail map in miles, the camp is at $(-3, -6)$, the lake at $(2, 6)$ and the trailhead at $(-1, 0)$. How far is the camp from the lake, in a straight line? → ***13 mi*** *The trailhead at (−1, 0) is not needed. 17 — added the legs [MA.8.GR.1.2]; a distance to the trailhead — the wrong pair of points*

**32–33. Board 9.** Mia says the segment from (0, 0) to (3, 1) is 3 units long, because it crosses 3 grid squares. Is she right? Explain, and find its length to the nearest tenth. → ***No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.*** *'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]; 4 — added the legs [MA.8.GR.1.2]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: a right triangle's perimeter, whole then to the tenth; 3 has no right angle and is split by its height, not in half; 4 votes on the perimeter errors. The pattern breaks at 5 — a map, and how much shorter the straight line is — then a rectangle's diagonal voted (6), a park map to the tenth (7), a map with an extra point (8); 9 is written. The independent set: a perimeter whole, a perimeter to the tenth, a parallelogram, straight against the streets, a map to the tenth, a rectangle's missing corner and its diagonal.

## Misconceptions to Watch

**Board 1.** 17 — left the slanted side out [Notes I: every side]

**Board 2.** 13 — left the slanted side out [Notes I]

**Board 3.** 40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]

**Board 4.** 4 + 7: left the slanted side out [Notes I: every side]

**Board 5.** 13 or 17 — one path, not how much shorter [the question asks a difference]

**Board 6.** 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]

**Board 7.** 14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]

**Board 8.** 17 — added the legs [MA.8.GR.1.2]

**Board 9.** 'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]

**Materials** Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.